

Lab 5 ticket

Why Did the System Behave That Way? — Friday, November 20

Name: _____ Language (circle): Octave / Python

New syntax (fill as you go)

I needed to...	Command / pattern (your language)
Eigenpairs of A	
k -th eigenvalue / vector	
Check $Av = \lambda v$	
Dot product $v_1^T v_2$	

Quiz 9 today is Week 11 inverses / rank / determinants (**not** this lab). Eigen fluency appears on Quiz 10.

E1. Lab 4 mystery: $B = \text{diag}(1.2, 0.5)$. Eigenpairs; why $(1, 0)$ grows and $(0, 1)$ decays.

λ, v : _____

One-sentence why: _____

E2. $A = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$. Eigenpairs; residual near 0? Dominant $|\lambda|$ and long-run direction.

λ 's: _____ Residuals OK? Y / N Dominant dir: _____

Expected: $\lambda = 1$ and 0.7 ; dominant along $(2, 1)$.

E3. Same A , $x_0 = (1000, 0)$; ~ 20 steps. 2–3 sentence eigen-explanation of stabilization:

E4. $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Software λ : _____ $x_0 \dots x_4$: _____

Complex $\lambda \leftrightarrow$ cycling (one sentence): _____

E5. Symmetric $S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. $v_1^T v_2 \approx$ _____ Orthogonal? Y / N

Stuck on (optional): E_____

Instructor initials: _____